

Katherina Teodorovich

EDUCATION

Purdue University | B.S. Computer Science, B.S. Mathematics

May 2025

GPA 4.0/4.0

- Relevant Coursework: Artificial Intelligence, Compilers, Analysis of Algorithms, Systems Programming, Data Structures & Algorithms, Computer Architecture
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EXPERIENCE

Software Engineer | *Airtable – San Francisco, CA*

Aug 2025 - Present

- Drove redesign of asynchronous task architecture, bringing a 20% decrease in polling query volume and 90% decrease in latency.
- Led design and implementation of persistence layer for AgentCore-sandboxed AI assistant
- Mentored a new college graduate

Undergraduate Teaching Assistant | *Purdue University – West Lafayette, IN*

Aug 2024 - May 2025

- UTA for CS 381: Analysis of Algorithms
- Assisted 100+ students in developing and proving algorithms of various archetypes during office hours

ML Software Intern | *DeGirum Corp. – Santa Clara, CA*

May 2024 - July 2024

- Optimized real-time object detection on RTSP Raspberry Pi camera feed to run at >30 FPS
- Doubled pose detection speed by refactoring C++ detection result postprocessing
- Wrote platform-agnostic benchmarking scripts for inference on Coral Edge TPU, Intel NPU, and CPU

ML Software Intern | *DeGirum Corp. – Santa Clara, CA*

May 2023 - Aug 2023

- Improved quantized YOLOv8 performance by 7% mAP by adjusting model architecture
- Implemented C++ postprocessor for YOLO detection, running at >100 FPS to parallelize with on-device inference
- Trained detection models for 5 tasks, and exported to 160 total model variants
 - Formats: yolov5nu/v5su/v8n/v8s, with SiLU/ReLU6 activations, to ONNX/TFLite/N2X

Undergraduate Data Science Researcher | *Battelle & The Data Mine – Purdue University*

Aug 2022 - Dec 2022

- Developed scaleable NLP hyperparameter tuning script using HyperBand algorithm
- Balanced runtime and performance of Random forest, K-means, and Hierarchical Clustering tuning algorithms

ML Software Intern | *DeGirum Corp. – Santa Clara, CA*

Jun 2022 - Aug 2022

- Automated Python label cleaning script for issues in training datasets such as Google's OpenImages
 - Removed double labels and flagged incorrect labels by comparing ground truth to inference results
 - Used K-fold cross validation to avoid model overfitting issues on small datasets
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EXTRACURRICULARS

Autonomy Team Member | *Autonomous Robotics Club – Purdue University*

Aug 2022 - Dec 2022

- Trained remote controlled cars with RL to play Rocket League (soccer) using Stable-Baselines3 and ROS

VP of Software | *FIRST Robotics Competition – Fremont High School*

Aug 2018 - May 2022

- Developed an auto-tracking turret mechanism that successfully shot 80% of balls into goal
 - Used custom OpenCV vision pipeline to detect retro-reflective field targets
 - Accounted for robot momentum by implementing feed-forward control on data from 4 rotary encoders on robot drivetrain
 - Used PID feedback control systems to optimally map joystick input data to motor voltages
 - Integrated motion profiling to optimize robot movement through predetermined paths
 - Managed a software development team of 9 programmers, maintaining the team's GitHub repo, oversaw all merges
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PROJECTS

ML Framework (github.com/kteodorovich/ml_framework)

Aug 2023 - Present

- Implemented ONNX operations (Conv2d, BatchNorm, AvgPool) in pure NumPy code
- Standalone ML framework with no dependencies on PyTorch, ONNXRuntime, or other runtime agents

YOLO Inference (github.com/DeGirum/yolo_inference)

Aug 2023

- Designed lightweight pre/post processing (letterbox padding, NMS, confidence thresholding) for YOLOv8
 - Made compatible with ONNX or TFLite, on CPU or EdgeTPU, through transposing tensors to appropriate input format
 - Evaluated exported model performance (mAP) on given dataset through pycocotools
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SKILLS

- *Languages*: Python, C/C++, Java, Typescript, Bash, SQL, Assembly
- *Tools/Frameworks*: VS Code, Cursor, Claude Code, JetBrains IDEs, Jupyter notebooks, Vim, Git, Github, SSH, GDB, PyTorch, ONNX, TFLite, Datadog, OpenSearch, Spinnaker